

GRC102: Information Security Governance

Week 4 Laboratory Assignment

Linux Auditing, Logging, Security Assessment and Control Monitoring

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Executive Summary

This laboratory exercise focused on implementing and evaluating Linux monitoring and auditing controls to support governance, risk management, and compliance objectives. The practical activities involved configuring and validating Auditd, reviewing system and authentication logs, conducting a security assessment using Lynis, and translating technical findings into governance-focused control monitoring activities.

The exercise demonstrated how security monitoring solutions can provide visibility into system activities, support accountability, assist in incident investigations, and improve overall security posture. Audit records, log analysis findings, and vulnerability assessment results were used to evaluate the effectiveness of existing controls and identify areas requiring remediation.

The results confirmed that continuous monitoring plays a critical role in ensuring compliance, detecting suspicious activity, and enabling informed governance decisions.

1. Scope and Authorization

1.1 Scope

The scope of this assessment included:

- Linux audit configuration using Auditd.
- Authentication log monitoring.
- System log analysis.
- Event reporting and investigation.

- Security assessment using Lynis.
- Control monitoring and governance evaluation.
- Mapping technical monitoring activities to SIEM and continuous monitoring processes.

1.2 Purpose

The purpose of the assessment was to evaluate the effectiveness of monitoring controls within a Linux environment and demonstrate how technical security evidence supports governance, risk management, and compliance activities.

2. Methodology

The following methodology was followed during the laboratory exercise:

Step 1: Audit Configuration

Auditd was verified, configured, and tested using custom monitoring rules to capture sensitive system events.

Step 2: Log Collection and Analysis

Linux logs were reviewed using:

- journalctl
- auth.log
- syslog

Relevant security events were identified and analyzed.

Step 3: Security Assessment

Lynis was used to perform a system security audit and identify weaknesses, warnings, and hardening opportunities.

Step 4: Governance Evaluation

Technical findings were translated into governance-focused control monitoring activities, remediation recommendations, and risk management actions.

3. Module 1: Auditd Configuration and Monitoring

3.1 Verification of Auditd Service

The Auditd service was verified to ensure it was installed and actively running on the Linux system.

```
(icdfa@icdfa)-[~]
$ sudo nano /etc/audit/rules.d/custom.rules

(icdfa@icdfa)-[~]
$ sudo systemctl restart auditd

(icdfa@icdfa)-[~]
$ sudo auditctl -l
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -F key=program_execution
-a always,exit -F arch=b32 -S execve -F key=program_execution
-w /var/log/auth.log -p wa -k auth_failures

(icdfa@icdfa)-[~]
$
```

Observation

The service status confirmed that Auditd was operational and capable of collecting audit records.

Governance Significance

Auditd supports accountability and traceability by recording privileged activities and critical system actions. This capability assists organizations in meeting compliance requirements and conducting forensic investigations.

3.2 Audit Rule Configuration

Custom monitoring rules were implemented to monitor:

- Modifications to /etc/passwd
- Modifications to /etc/shadow
- Program execution events
- Authentication-related log activity

```

time→Fri Oct  2 21:18:40 2026
type=PROCTITLE msg=audit(1790972320.008:1112): proctitle=2F62696E2F7368002F7573722F73686172652
F6B616C692D7468656D65732F78666365342D70616E656C2D67656E6D6F6E2D76706E69702E7368
type=PATH msg=audit(1790972320.008:1112): item=3 name="/lib64/ld-linux-x86-64.so.2" inode=7929
12 dev=08:01 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0
cap_fver=0 cap_frootid=0
type=PATH msg=audit(1790972320.008:1112): item=2 name="/bin/sh" inode=786236 dev=08:01 mode=01
00755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_froot
id=0
type=PATH msg=audit(1790972320.008:1112): item=1 name="/usr/share/kali-themes/xfce4-panel-genm
on-vpnip.sh" inode=1076068 dev=08:01 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap
_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1790972320.008:1112): item=0 name="/usr/share/kali-themes/xfce4-panel-genm
on-vpnip.sh" inode=1076068 dev=08:01 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap
_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790972320.008:1112): cwd="/home/icdfa"
type=EXECVE msg=audit(1790972320.008:1112): argc=2 a0="/bin/sh" a1="/usr/share/kali-themes/xfce4-panel-genmon-vpnip.sh"
type=SYSCALL msg=audit(1790972320.008:1112): arch=c000003e syscall=59 success=yes exit=0 a0=55
6c48e3e2d0 a1=556c48e460d0 a2=7ffffae72d7e8 a3=0 items=4 ppid=1456 pid=11731 auid=1000 uid=1000
gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000 tty=(none) ses=2 comm=
"xfce4-panel-gen" exe="/usr/bin/dash" subj=unconfined key="program_execution"
time→Fri Oct  2 21:18:40 2026
type=PROCTITLE msg=audit(1790972320.024:1113): proctitle=69700074756E7461700073686F77
type=PATH msg=audit(1790972320.024:1113): item=2 name="/lib64/ld-linux-x86-64.so.2" inode=7929

```

Observation

The configured rules were successfully loaded and displayed through Auditd.

Governance Significance

Monitoring privileged files and command execution helps establish accountability, detect unauthorized activities, and support audit requirements.

3.3 Event Generation and Validation

Test activities were performed to generate audit records.

Events generated included:

- Access to /etc/passwd
- Command execution activities
- Authentication-related events

Audit records were reviewed using:

- ausearch
- Aureport

```
File Actions Edit View Help

(icdfa@icdfa)-[~]
$ sudo systemctl status auditd
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-10-02 21:09:55 WAT; 34s ago
     Invocation: 01c981ed1d4b4d968a5dd788bff3d228
       Docs: man:auditd(8)
             https://github.com/linux-audit/audit-documentation
    Main PID: 7140 (auditd)
      Tasks: 2 (limit: 4545)
     Memory: 660K (peak: 1.8M)
        CPU: 25ms
      CGroup: /system.slice/auditd.service
             └─7140 /usr/sbin/auditd

Oct 02 21:09:55 icdfa systemd[1]: Starting auditd.service - Security Audit Logging Service...
Oct 02 21:09:55 icdfa auditd[7140]: No plugins found, not dispatching events
Oct 02 21:09:55 icdfa auditd[7140]: Init complete, auditd 4.1.2 listening for events (startup state enabled)
Oct 02 21:09:55 icdfa systemd[1]: Started auditd.service - Security Audit Logging Service.
... skipping ...
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-10-02 21:09:55 WAT; 34s ago
     Invocation: 01c981ed1d4b4d968a5dd788bff3d228
       Docs: man:auditd(8)
             https://github.com/linux-audit/audit-documentation
    Main PID: 7140 (auditd)
      Tasks: 2 (limit: 4545)
     Memory: 660K (peak: 1.8M)
        CPU: 25ms
      CGroup: /system.slice/auditd.service
             └─7140 /usr/sbin/auditd

sudo aureport --login
Home
Summary Report
=====
Range of time in logs: 02/10/2026 21:09:55.322 - 02/10/2026 21:20:26.261
Selected time for report: 02/10/2026 21:09:55 - 02/10/2026 21:20:26.261
Number of changes in configuration: 11
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 3
Number of terminals: 7
Number of host names: 2
Number of executables: 18
Number of commands: 19
Number of files: 31
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 14
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 4
Number of process IDs: 1606
Number of events: 1834
```

Findings

Auditd successfully captured monitored events and generated corresponding records.

Interpretation

The collected audit records demonstrate that monitoring controls are functioning effectively and provide reliable evidence for investigations, compliance reporting, and control assurance activities.

4. Module 2: Linux Log Analysis

4.1 Journal Analysis

System events were reviewed using Journalctl.

Commands reviewed included:

- `journalctl`
- `journalctl -u ssh`
- `journalctl --since today`
- `journalctl -p err`

```
File Actions Edit View Help

Failed Summary Report
=====
Range of time in logs: 02/10/2026 21:09:55.322 - 02/10/2026 21:20:26.365
Selected time for report: 02/10/2026 21:09:55 - 02/10/2026 21:20:26.365
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 1
Number of terminals: 2
Number of host names: 0
Number of executables: 2
Number of commands: 2
Number of files: 11
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 14
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 2
Number of process IDs: 3
Number of events: 14

Login Report
=====
# date time audit host term exe success event
=====
<no events of interest were found>

(icdfa@icdfa)-[~]
$
```

Findings

Observed events included:

- System service activity
- Login events
- Warning and error messages

Interpretation

Journal logs provide centralized visibility into operating system activities and support operational monitoring.

4.2 Authentication Log Analysis

Authentication logs were reviewed using:

```
1 /var/log/auth.log
```

The review focused on:

- Login attempts

- Failed authentications
- Sudo activities

```
(icdfa@icdfa)-[~]
$ sudo journalctl
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu->
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000007ffeffff] usable
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0000000007ffffffffff] ACPI data
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/>
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
May 10 20:50:54 icdfa kernel: Hypervisor detected: KVM
May 10 20:50:54 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
May 10 20:50:54 icdfa kernel: kvm-clock: using sched offset of 155397490851 cycles
May 10 20:50:54 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x>
May 10 20:50:54 icdfa kernel: tsc: Detected 2419.196 MHz processor
May 10 20:50:54 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
May 10 20:50:54 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
May 10 20:50:54 icdfa kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000
May 10 20:50:54 icdfa kernel: MTRR map: 4 entries (3 fixed + 1 variable; max 35), built from >
May 10 20:50:54 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
May 10 20:50:54 icdfa kernel: found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
May 10 20:50:54 icdfa kernel: RAMDISK: [mem 0x282fb000-0x30174fff]
May 10 20:50:54 icdfa kernel: ACPI: Early table checksum verification disabled
May 10 20:50:54 icdfa kernel: ACPI: RSDP 0x000000000000E000 000024 (v02 VBOX )
May 10 20:50:54 icdfa kernel: ACPI: XSDT 0x000000007FFF0030 00003C (v01 VBOX VBOXXSDT 00000>
May 10 20:50:54 icdfa kernel: ACPI: FACP 0x000000007FFF00F0 0000F4 (v04 VBOX VBOXFACP 00000>
May 10 20:50:54 icdfa kernel: ACPI: DSDT 0x000000007FFF0610 002353 (v02 VBOX VBOXBIOS 00000>
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: FACS 0x000000007FFF0200 000040
May 10 20:50:54 icdfa kernel: ACPI: APIC 0x000000007FFF0240 000054 (v02 VBOX VBOXAPIC 00000>
```

```
(icdfa@icdfa)-[~]
$ sudo journalctl --since "today"
Oct 02 21:00:46 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu->
Oct 02 21:00:46 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=>
Oct 02 21:00:46 icdfa kernel: BIOS-provided physical RAM map:
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000000dfffff] usable
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x0000000000dfff0000-0x0000000000dfffffff] ACPI data
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Oct 02 21:00:46 icdfa kernel: BIOS-e820: [mem 0x0000000100000000-0x000000011fffffff] usable
Oct 02 21:00:46 icdfa kernel: NX (Execute Disable) protection: active
Oct 02 21:00:46 icdfa kernel: APIC: Static calls initialized
Oct 02 21:00:46 icdfa kernel: SMBIOS 2.5 present.
Oct 02 21:00:46 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/
Oct 02 21:00:46 icdfa kernel: DMI: Memory slots populated: 0/0
Oct 02 21:00:46 icdfa kernel: Hypervisor detected: KVM
Oct 02 21:00:46 icdfa kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Oct 02 21:00:46 icdfa kernel: kvm-clock: using sched offset of 35973183358 cycles
Oct 02 21:00:46 icdfa kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x
Oct 02 21:00:46 icdfa kernel: tsc: Detected 2112.006 MHz processor
Oct 02 21:00:46 icdfa kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Oct 02 21:00:46 icdfa kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Oct 02 21:00:46 icdfa kernel: last_pfn = 0x120000 max_arch_pfn = 0x400000000
Oct 02 21:00:46 icdfa kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from
Oct 02 21:00:46 icdfa kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Oct 02 21:00:46 icdfa kernel: CPU MTRRs all blank - virtualized system.
Oct 02 21:00:46 icdfa kernel: last_pfn = 0xe0000 max_arch_pfn = 0x400000000
Oct 02 21:00:46 icdfa kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
Oct 02 21:00:46 icdfa kernel: RAMDISK: [mem 0x280b1000-0x3004ffff]
Oct 02 21:00:46 icdfa kernel: ACPI: Early table checksum verification disabled
Oct 02 21:00:46 icdfa kernel: ACPI: RSDP 0x000000000000E000 000024 (v02 VBOX )
```

```
File Actions Edit View Help
(icdfa@icdfa)-[~]
$ sudo journalctl -p err
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running o>
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely>
May 10 20:50:54 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported>
May 10 20:57:08 icdfa lightdm[761]: gkr-pam: unable to locate daemon control file
May 10 20:57:08 icdfa lightdm[645]: pam_systemd(lightdm-greeter:session): Failed to release s>
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registry: Error calling StartServiceByN>
May 10 20:57:21 icdfa obexd[1254]: Unable to acquire registry: Error calling StartServiceByN>
-- Boot f8a18b2290a640c9b9236a0ff3f2a536 --
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running o>
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely>
May 11 04:45:41 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported>
May 11 04:51:38 icdfa lightdm[802]: gkr-pam: unable to locate daemon control file
May 11 04:51:38 icdfa lightdm[690]: pam_systemd(lightdm-greeter:session): Failed to release s>
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registry: Error calling StartServiceByN>
May 11 04:51:48 icdfa obexd[1324]: Unable to acquire registry: Error calling StartServiceByN>
-- Boot 1330474cb531439e929342180494b74a --
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running o>
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely>
May 11 06:10:01 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported>
May 11 06:10:30 icdfa lightdm[785]: gkr-pam: unable to locate daemon control file
May 11 06:10:30 icdfa lightdm[678]: pam_systemd(lightdm-greeter:session): Failed to release s>
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registry: Error calling StartServiceByN>
May 11 06:10:42 icdfa obexd[1293]: Unable to acquire registry: Error calling StartServiceByN>
-- Boot b8f421d015a94e7d82bf378d620f3ea1 --
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running o>
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely>
May 11 06:22:39 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported>
May 11 06:23:02 icdfa lightdm[741]: gkr-pam: unable to locate daemon control file
May 11 06:23:02 icdfa lightdm[635]: pam_systemd(lightdm-greeter:session): Failed to release s>
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registry: Error calling StartServiceByN>
May 11 06:23:17 icdfa obexd[1256]: Unable to acquire registry: Error calling StartServiceByN>
-- Boot 70de93df416a4f15a4e11555f8166985 --
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running o>
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely>
May 11 06:29:53 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported>
May 11 06:30:16 icdfa lightdm[740]: gkr-pam: unable to locate daemon control file
```

Findings

Observed security-relevant events included:

- Successful user authentication
- Sudo command execution
- Failed authentication attempts

Risk Analysis

Repeated failed login attempts may indicate:

- Password guessing attacks
- Misconfigured services
- Unauthorized access attempts

Governance Recommendation

Organizations should establish thresholds for failed authentication events and trigger alerts whenever unusual activity is detected.

4.3 System Log Analysis

System logs were reviewed through:

```
1 /var/log/syslog
```

```
(icdfa@icdfa)-[~]
$ sudo less /var/log/auth.log
/var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "failed password" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ sudo grep -i "sudo" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
```

Findings

Observed system messages included:

- Service notifications
- Warning entries
- Error messages

Interpretation

Continuous review of system logs assists administrators in detecting operational issues before they escalate into security incidents.

4.4 Event Analysis Table

Event	Log Source	Security Significance	Recommended Action
Sudo Command Execution	auth.log	Privileged access activity	Periodic privileged-user review
Failed Login Attempt	auth.log	Potential unauthorized access attempt	Investigate repeated failures
Service Error	syslog	Operational risk indicator	Review service configuration
Program Execution Event	Auditd	User activity monitoring	Maintain auditing controls
File Access Event	Auditd	Sensitive file monitoring	Continue continuous monitoring

5. Module 3: Lynis Security Assessment

5.1 Security Audit Execution

Lynis was executed to perform a comprehensive system assessment.

Evidence: Appendix C - Screenshot 8

Objective

To identify security weaknesses, misconfigurations, and hardening opportunities.

5.2 Assessment Findings

The assessment generated a hardening score and several recommendations.

File Actions Edit View Help

```
(icdfa@icdfa)-[~]  
$ lynis show version  
3.1.6
```

```
(icdfa@icdfa)-[~]  
$ sudo lynis audit system
```

[Lynis 3.1.6]

```
#####  
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are  
welcome to redistribute it under the terms of the GNU General Public License.  
See the LICENSE file for details about using this software.
```

```
2007-2025, CISOfy - https://cisofy.com/lynis/  
Enterprise support available (compliance, plugins, interface and tools)
```

```
#####
```

[+] **Initializing program**

```
- Detecting OS... [ DONE ]  
- Checking profiles... [ DONE ]
```

```
Program version:      3.1.6  
Operating system:     Linux  
Operating system name: Kali Linux  
Operating system version: Rolling release  
End-of-life:          UNKNOWN  
Kernel version:       6.12.13  
Hardware platform:    x86_64  
Hostname:             icdfa
```

```
Profiles:             /etc/lynis/default.prf  
Log file:              /var/log/lynis.log  
Report file:          /var/log/lynis-report.dat  
Report version:       1.0
```

- Plugin: **debian**

[

[+] **Debian Tests**

- Checking for system binaries that are required by Debian Tests...

- Checking /bin... [**FOUND**]
- Checking /sbin... [**FOUND**]
- Checking /usr/bin... [**FOUND**]
- Checking /usr/sbin... [**FOUND**]
- Checking /usr/local/bin... [**FOUND**]
- Checking /usr/local/sbin... [**FOUND**]

- Authentication:

- PAM (Pluggable Authentication Modules):

[**WARNING**]: Test DEB-0001 had a long execution: 86.044523 seconds

- libpam-tmpdir [**Not Installed**]

- File System Checks:

- DM-Crypt, Cryptsetup & Cryptmount:
- Checking / on /dev/sda1 [**NOT ENCRYPTED**]

- Software:

- apt-listbugs [**Not Installed**]
- apt-listchanges [**Not Installed**]
- needrestart [**Not Installed**]
- fail2ban [**Not Installed**]

]

[+] **Boot and services**

- Service Manager [**systemd**]
- Checking UEFI boot [**DISABLED**]
- Checking presence GRUB2 [**FOUND**]
- Checking for password protection [**NONE**]
- Check running services (systemctl) [**DONE**]
Result: found 19 running services
- Check enabled services at boot (systemctl) [**DONE**]
Result: found 20 enabled services

```

- Check startup files (permissions) [ OK ]

[WARNING]: Test BOOT-5184 had a long execution: 16.093035 seconds

- Running 'systemd-analyze security'
Unit name (exposure value) and predicate

- ModemManager.service (value=6.3) [ MEDIUM ]
- NetworkManager.service (value=7.8) [ EXPOSED ]
- accounts-daemon.service (value=5.5) [ MEDIUM ]
- auditd.service (value=9.4) [ UNSAFE ]
- colord.service (value=3.5) [ PROTECTED ]
- cron.service (value=9.6) [ UNSAFE ]
- dbus.service (value=9.3) [ UNSAFE ]
- dm-event.service (value=9.5) [ UNSAFE ]
- emergency.service (value=9.5) [ UNSAFE ]
- getty@tty1.service (value=9.6) [ UNSAFE ]
- haveged.service (value=3.2) [ PROTECTED ]
- lightdm.service (value=9.6) [ UNSAFE ]
- lvm2-lvmpolld.service (value=9.5) [ UNSAFE ]
- lynis.service (value=9.6) [ UNSAFE ]
- pcscd.service (value=2.1) [ PROTECTED ]
- plymouth-start.service (value=9.5) [ UNSAFE ]
- polkit.service (value=1.2) [ PROTECTED ]
- rc-local.service (value=9.6) [ UNSAFE ]
- rescue.service (value=9.5) [ UNSAFE ]
- rtkit-daemon.service (value=7.2) [ MEDIUM ]
- smartmontools.service (value=9.6) [ UNSAFE ]
- ssh.service (value=9.6) [ UNSAFE ]
- systemd-ask-password-console.service (value=9.4) [ UNSAFE ]
- systemd-ask-password-plymouth.service (value=9.5) [ UNSAFE ]
- systemd-ask-password-wall.service (value=9.4) [ UNSAFE ]
- systemd-bsod.service (value=9.5) [ UNSAFE ]
- systemd-hostnamed.service (value=1.7) [ PROTECTED ]
- systemd-initctl.service (value=9.4) [ UNSAFE ]
- systemd-journald.service (value=4.9) [ PROTECTED ]
- systemd-logind.service (value=2.8) [ PROTECTED ]
- systemd-networkd.service (value=2.9) [ PROTECTED ]

```

Interpretation

The Lynis results indicate that while baseline security controls are present, additional hardening measures are needed to reduce risk and strengthen compliance with security best practices.

6. Control Monitoring and Governance Evaluation

The technical evidence collected during the laboratory exercise was mapped to governance and control monitoring activities.

Control Objective	Evidence Source	Control Owner	Status	KPI / KRI	Risk Rating	Remediation
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Monitor privileged activities	Auditd logs	System Administrator	Effective	Unauthorized changes detected	Medium	Continue monitoring
Detect failed logins	auth.log	Security Administrator	Effective	Failed logins per day	High	Investigate anomalies
Monitor system health	journalctl	Operations Team	Effective	Critical errors	Medium	Resolve recurring issues
Assess system security posture	Lynis	Security Team	Partially Effective	Critical findings	High	Implement recommendations
Maintain audit accountability	Audit reports	Compliance Team	Effective	Audit coverage rate	Medium	Review periodically

7. SIEM and Continuous Monitoring Mapping

Modern organizations consolidate security information through Security Information and Event Management (SIEM) platforms such as Microsoft Sentinel, Splunk, and IBM QRadar.

Data Source	SIEM Capability	Governance Escalation Trigger
Auditd	Audit event collection	Unauthorized file modifications
auth.log	Authentication monitoring	Multiple failed login attempts
journalctl	Event correlation	Service outages or recurring errors
Lynis	Vulnerability management	Critical security findings

Governance Value

Integration of security logs into a SIEM platform provides:

- Real-time monitoring.
- Centralized visibility.
- Automated alert generation.
- Compliance reporting.
- Faster incident response.

Continuous monitoring enables management to make informed risk-based decisions and ensures security controls remain effective.

8. Remediation and Retest Plan

Finding	Recommended Action	Responsible Owner	Retest Method
Weak security configuration	Apply Lynis recommendation	Linux Administrator	Re-run Lynis audit
Authentication anomalies	Review account activity	IAM Administrator	Monitor authentication logs
Service warnings/errors	Investigate root cause	System Administrator	Validate normal operation
Control monitoring gaps	Expand monitoring coverage	Security Team	Review audit reports

9. Conclusion

The laboratory exercise successfully demonstrated the implementation and evaluation of Linux auditing and monitoring controls. Auditd, Linux system logs, and Lynis provided valuable security information that can be used to support governance, risk management, compliance monitoring, and incident investigations.

The findings confirmed the importance of continuous monitoring as a foundational component of an effective security governance programme. By converting technical monitoring outputs into governance-focused control assessments, organizations can improve accountability, strengthen compliance, and enhance overall cyber resilience.